

with sample rates up to 192ksps. It is used to measure different analogue and battery inputs.

Implementation of touchscreen based design on the front panel instead of an LCD display and keypad provides better user interface and flexibility.

External devices. PSoC can communicate with external devices like iPod and iPhone through UART and USB protocol. You can control such devices.

PSoC4 supports code security for home appliance solutions due to programmable architecture. A PSoC can connect to devices like Bluetooth controllers, so that you can create a network of automated devices.

In the world of cortex processors, operating range is up to 3.3V. PSoC4 supports an operating range of 1.71V to 5.5V.

Failure analysis and returned materials. Increasing the number of internal and external interfaces on the board will increase the number of ways that an intruder can create havoc on the system. This is one of the single-largest limitations of this embedded system, which can be solved by a single-chip solution.

A PSoC has an internal RTC component for real-time measurement. It does not require external clock/oscillator circuitry.

It supports USB 2.0 interface, allowing you to interface external memory (hard disk) through USB 2.0. It also supports SD card interface. A PSoC has an internal 8-bit digital-to-analogue converter (DAC), which can be used for controlling speaker volume and for muting the speaker.

Design challenges

An MCU with one-time programmable feature prevents reverse engineering of the firmware by competitors and hackers. Implementing protection against voltage fluctuations, waterproofing and water-tolerance

is a design challenge for the system designer. The solution involves electromechanical construction and designing a compact and cost-effective electromechanical solution.

Certifying this electromechanical design with EMI/EMC standards is another challenge for the designer. Using flexible printed circuits interface for displays, touch panels and buttons is also a design challenge.

A PSoC supports CapSense technology. Using PSoC in the design, CapSense based button, slider and proximity can be implemented for front-panel design. Meeting CapSense performance (signal-to-noise ratio) with nearby LEDs (PWM based) on the front panel is also a challenge for the system designer.

Implementing auto-tuning with variation in trace capacitance, variation in CapSense button and slider sizes and shape are other design challenges, along with implementing CapSense sensing with thicker glass material (display glass) and meeting CapSense sensitivity with that type of materials.

Fault detection and recovery mechanism is required in most application. Power supply design with battery protection, over-current, over-heating, start-up fail condition is required for home appliance applications. Implementing self-diagnostics in home appliance applications is another design challenge for the system designer.

Last, but not the least, as final solution is expected to work 24/7 continuously, component selection and reliability is a challenge.

Conclusion

A PSoC is a combination of an MCU and ASIC. It provides ease-of-use environment. Using PSoC in above-discussed applications helps reduce product cost (by reducing bills of materials cost) and project cost (with PSoC Creator and PSoC Designer implementation). **EFY**

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INFRARED THERMOMETERS



IRT1050P

- Temp. Range: -50°C ~ 1050°C
- Accuracy: $\pm 1.5^\circ\text{C}$
- Distance Spot Ratio: 50 : 1
- Emissivity: 0.10 ~ 1.00 (Adjustable)

IRT550P

- Temp. Range: -50°C ~ 550°C
- Accuracy: $\pm 1.5^\circ\text{C}$
- Distance Spot Ratio: 12 : 1
- Emissivity: 0.95

IRT380P

- Temp. Range: -50°C ~ 380°C
- Accuracy: $\pm 1.5^\circ\text{C}$
- Distance Spot Ratio: 12 : 1
- Emissivity: 0.95

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